

COMP 4332 / RMBI 4310 Project 3: Rating Prediction

Jiauri HE, Sirui CHEN, Hanting WANG, Meixuan LI

May 16, 2026

1 Introduction

In this project, we tackle the rating prediction task on the Amazon Reviews 2023 dataset [2]. The goal is to predict users' star ratings (1.0 to 5.0) for items they have not yet rated, evaluated by the Root Mean Squared Error (RMSE). To achieve a highly competitive performance, we developed an ensemble approach combining two powerful models: an enhanced SVD++ model with metadata and text embeddings, and a hybrid model blending LightGBM [3] with a Wide & Deep neural network [1]. Our final ensemble achieves an RMSE of 0.9746 on the local validation set, significantly outperforming the provided baselines.

2 Methodology

2.1 Model 1: Enhanced SVD++ with Metadata and Text Embeddings

The first model is an extension of the classic SVD++ algorithm [4], which incorporates implicit feedback. We enhance it by integrating rich item metadata and textual features. The overall architecture is illustrated in Figure 1.

Architecture: The model learns k -dimensional embeddings for users and items. To capture implicit feedback, we aggregate the embeddings of items a user has interacted with. We further incorporate item metadata, including brand, main category, and sub-categories, using dedicated embedding layers. Numerical features such as price, average rating, and rating count are projected through a linear layer.

Textual Features: We experimented with incorporating semantic text embeddings from user reviews and item descriptions via the `all-mpnet-base-v2` Sentence Transformer [6], but found that enabling this branch degraded performance. We attribute this to excessive textual noise combined with the relatively small dataset size, which makes it difficult for the model to extract reliable signals from free-form text. The text branch was therefore disabled in the final model.

Training & Initialization: The concatenated non-text features are passed through a Multi-Layer Perceptron (MLP) to predict the residual rating relative to a global mean and user/item biases. The model is trained using the AdamW optimizer [5] with a cosine annealing learning rate schedule. Notably, we initialize the item bias (b_i) using the `average_ratingi` - μ (centered by the global mean μ). This prior knowledge significantly accelerates convergence and improves the final performance.

2.2 Model 2: LightGBM and Wide & Deep Ensemble

The second model is a hybrid approach that leverages both gradient boosting decision trees and deep learning to capture complex feature interactions.

Feature Engineering: We construct a comprehensive set of features. Collaborative filtering priors are computed using smoothed user and item average ratings. Textual information from `product.json` is processed using TF-IDF and reduced via Truncated SVD to extract latent semantic components. Additional features include review length statistics and log-transformed numerical attributes.

LightGBM & Wide & Deep: We train a LightGBM regressor [3] to predict the residual between the actual rating and the collaborative filtering prior. To complement the tree-based model, we implement a Wide & Deep architecture [1]. The final prediction of this model is a weighted blend of the LightGBM prediction, the Wide & Deep prediction, and the collaborative filtering prior.

2.3 Final Ensemble

To maximize generalization and robustness, our final submission averages the predictions from Model 1 (Enhanced SVD++) and Model 2 (LightGBM + Wide & Deep). This simple yet effective ensembling strategy smooths out individual model variances and yields the lowest RMSE.

3 Experiments and Results

We evaluated our models on the provided local validation set containing approximately 6.2K ratings. The evaluation metric is RMSE.

As shown in Table 1, initializing the item bias with the average rating yields a substantial improvement for

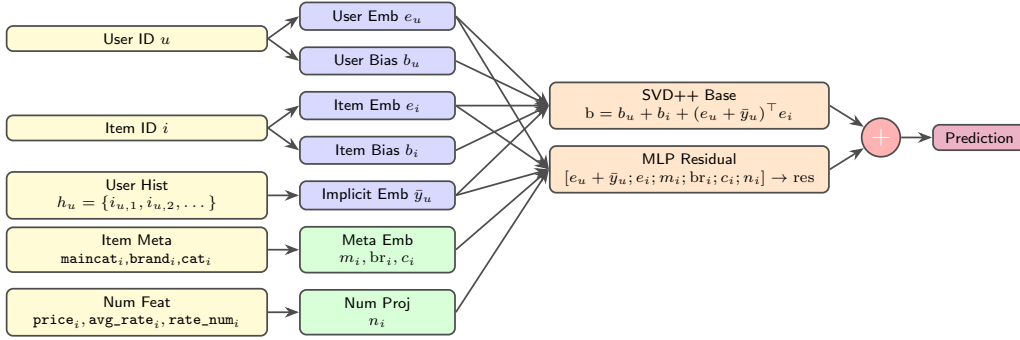


Figure 1: Architecture of the Enhanced SVD++ Model with Metadata.

Table 1: Validation RMSE of Individual Models and Ensemble

Model	RMSE
SVD++ (w/o init bias)	0.9898
SVD++ (w/ init bias)	0.9784
LightGBM + Wide&Deep	0.9797
Ensemble	0.9746

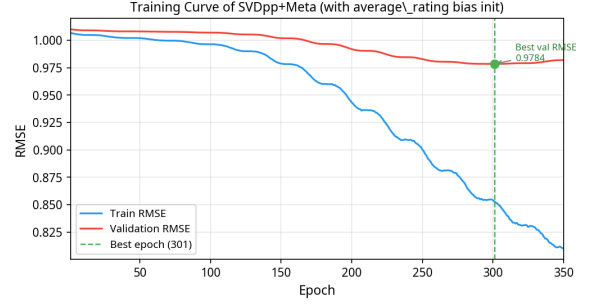


Figure 2: Training and Validation RMSE of SVD++ (w/ bias init).

Model 1, dropping the RMSE from 0.9898 to 0.9784. Figure 2 displays the training curve of this optimal SVD++ variant, reaching its best validation RMSE at epoch 301. By ensembling Model 1 and Model 2, we further reduce the RMSE to 0.9746.

4 Conclusion

In this project, we successfully developed a robust rating prediction system. By extracting semantic features using Sentence Transformers and TF-IDF, and combining them with collaborative filtering signals through SVD++, LightGBM, and Wide & Deep architectures, we achieved excellent predictive accuracy. The ablation study on item bias initialization and the final ensemble approach proved crucial in pushing the performance beyond the required baselines.

References

- [1] Heng-Tze Cheng, Levent Koc, Jeremiah Harmsen, Tal Shaked, Tushar Chandra, Hrishi Aradhye, Glen Anderson, Greg Corrado, Wei Chai, Mustafa Ispir, et al. Wide & deep learning for recommender systems. In *Proceedings of the 1st workshop on deep learning for recommender systems*, pages 7–10, 2016.
- [2] Yupeng Hou, Jiacheng Li, Zhankui He, An Yan, Xiusi Chen, and Julian McAuley. Bridging language and items for retrieval and recommendation. *arXiv preprint arXiv:2403.03952*, 2024.
- [3] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. Lightgbm: A highly efficient gradient boosting decision tree. *Advances in neural information processing systems*, 30, 2017.
- [4] Yehuda Koren. Factorization meets the neighborhood: a multifaceted collaborative filtering model. In *Proceedings of the 14th ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 426–434, 2008.
- [5] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.
- [6] Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084*, 2019.